ITEM – 1

MODULE 10

$A_t = A_0 e^{-\lambda t}$, $\lambda = \frac{In^2}{5730}\ s^{-1}$

$500 = 4000e^{\frac{In^2}{5730}t}$, $t = 17190$ years

Since **17190 years > 16900 years**, then the site should be preserved as part of the region's cultural heritage.

Wavelength of x rays

$$\text{from eV} = \frac{hc}{Amin}\ \lambda min, = \frac{6.6 \times 10^{-34} \times 3 \times 10^{8})}{1.6 \times 10^{-19} \times 80000}$$

$$\lambda_{min} = 1.55 \times 10^{-10}\ m$$

Since $1.55 \times 10^{-11}\ m < 1.6 \times 10^{-11}\ m$, the scanner provides accurate information.

Production of X-Rays

- When low voltage heats the filament cathode, electrons are emitted thermionically.
- Emitted electrons are accelerated by high voltage across the tube towards the metal target.
- On hitting the metal target, 99% of the kinetic energy of electrons is converted into heat and 1% of its kinetic energy is converted into x-rays.
- For continuous production of x-rays, excess heat generated at the metal target is conducted by the anode to the cooling fins where it is extracted.

b) A line spectrum is used to identify a metal because each metal emits / absorbs light at its own set of wavelengths.

- The produced x rays are directed to the metal penetrate deep and dislodges the electrons in the inner most shells.
- The excited electrons jump to higher energy levels and on their way to lower energy levels they emit wavelengths.
- This emitted light is passed through a prism / diffraction grating in a spectroscope producing a line spectrum.
- The positions of the spectral lines are compared with the known spectra of different metals.
- If the lines match those of a known metal, the metal is identified.

b)

1. Detecting broken bones
2. Testing welds, detecting cracks in metal
3. Determining time of death
4. To track how drugs move in the body

Explain any 2

- The photographic film is placed behind the suspected part of the body.
- X-rays are then directed to the suspected part where x-rays can pass through the broken part but are absorbed by unbroken bones.

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- The solid unbroken part will appear light but the broken parts will appear dark.

4) the photographic plate

- The drug is injected into the body, and inside the body it decays by emitting γ-rays (gamma-rays).
- Detectors around the body catch both gamma-rays at the same time, and the computer maps where they came from. Bright spots indicate where the drug has accumulated.

Item - 2

a) Being Pushed Sideways

- As the car moved round the bend along a circular path, it continuously changes direction which means it undergoes centripetal acceleration directed towards the center. Aisha and peter`s bodies possess inertia and naturally tend to maintain their original straight-line motion.
- And inside a turning car the passengers experience an outward friction force known as the centrifugal.

4

Why was the car not sliding?

- The static friction between the car tyre`s and road surface is strong enough to provide the required inward centripetal force.
- The passengers slide sideways relative to the car until the lateral support of the seats or doors applies an adequate inward normal force to pull them along the curve.

3

$$F_c \leq F_f,\ F_c = \frac{m \cdot \mu^2}{r},\quad F_f = \mu \cdot m \cdot g$$

Did Jose negotiate the corner at the right speed?

> *Given constants:*
> *r = 80 m, μ = 0.28, g = 9.8 ms⁻²*

$$(m \cdot v_max^2) / r \leq \mu mg \quad => \quad v_max = \sqrt{(\mu rg)}$$

Jose negotiating the corner at right speed?

R=80m, M = 0.28, g=9.81ms^{-1}

$\frac{MV^2max}{r} \leq \mu mg, Vmax = \sqrt{\mu rg}$

Vmax = $\sqrt{0.28 \times 9.81 \times 80} = 14.82 ms^{-1}$

6

Engineers recommend only 90% of this speed. V = 0.9x14.82 = 13.34ms^{-1} since Jose was travelling at 12ms^{-1} thus is below 13.34ms^{-1} he was negotiating the corner safely.

Page 4

Jose's Options at the Junction

> *Distance to clear junction: s = 50 + 20 = 70 m*
> *Time before red: t = 4 s*

Option 1: Accelerate at 1.4 ms⁻²

$s = u \cdot t + \frac{1}{2} \cdot a \cdot t^2$

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$s = 12(4) + ½(1.4)(4)^2 = 59.2\ m$

He requires 70 m. So 59.2 m < 70 m, he does not clear the junction, he will be stranded right in the intersection when traffic light turns red.

Option 2: Decelerate at 0.4 ms^{-2}

$s = 12(4) - ½(0.4)(4)^2 = 44.8\ m$

44.8 m < 50 m, the car will remain entirely behind the stop line when light turns red. Hence, he will safely avoid entering the intersection during red light.

Final Conclusion: option 2 is significantly safer option, and option 1 guarantees a dangerous traffic violation and potential collision inside the junction.

b) Safety Challenges on Bends & Junctions

- The wet road hazard; during rainy periods, water acts as a lubricant between the tyre`s and the road surface drastically lowering coefficient of friction which reduces the maximum speed safe speed, causing normal drivers moving at standard speeds to lose traction and skid off road.
- Unbanked design vulnerability; rely entirely on tyre`s road friction for centripetal force is fundamentally flawed because friction is highly unpredictably and weather dependent.

Technical Advice to the Authorities

To eliminate the black sport and stabilize the flow of traffic, the following engineering modifications are advised.

- Redesign the bend with mechanical banking, the authorities should rebuild the sharp bend by tilting the road surface inwards at a strategic angle. By banking the curve, the normal reaction force splits into horizontal component of normal reaction that points inwards the center of the turn which minimizes skidding accidents during wet periods.
- Apply high-friction surface treatment; Resurface the bend using highly durable coarse aggregates embedded in a polymer. This artificial texturing maintains a high coefficient of friction even when wet, preventing vehicles from sliding out.

Item - 3

a) Variation in Sound Intensity and Clarity of a Public Address System

- Variation in loudness is due to the interference of sound waves coming from the two loudspeakers.
- At some positions, sound waves meet in phase, overlap, and undergo constructive interference, producing a loud sound.
- At other positions, they arrive out of phase, cancel each other out, and undergo destructive interference.

Calculation Context: Isaac counted 9 loud sound positions over a specific distance.

Distance walked: $d = v \times t = 3 \times 8 = 24\ m$

Spacing between adjacent loud positions:

$Dx = \frac{24}{8} = 3.0\ m$

$Dx = \frac{v}{\lambda}, \lambda = 3x2 = 6m$

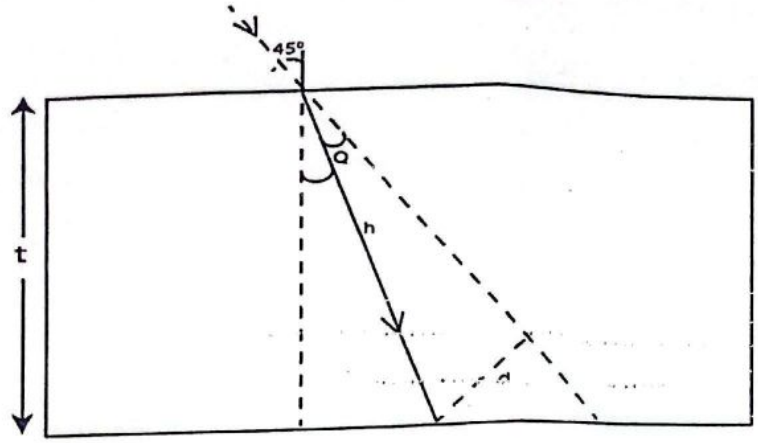
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$F = \frac{}{\lambda} = \frac{}{6}$ — 56.7 Hz

Conclusion: Since 56.7 Hz > 55 Hz,

The public address system would be clear because the amplifier frequency is above the required minimum threshold of 55 Hz.

b) Lateral Displacement Through Glass



Using Snell's Law nsini = constant:

$1.0 \times \sin(45°) = 1.50 \times \sin(r)$

$r = \sin^{-1} \frac{\sin 45°}{1.5} = 28.1°$

$\cos(r) = \frac{t}{h}$ $h = \frac{t}{\cos r}$

$\sin(\theta) = \frac{d}{h}$, $h = \frac{d}{\sin o}$

$\frac{t}{\cos 28} = \frac{d}{\sin 16.9^0}$ (16.9°)

$\frac{1.2}{\cos 28.1} = \frac{d}{\sin 16.9}$

$d = 0.39545$ cm

$d = 0.4$ cm

Since 0.4 cm < 0.5 cm, the lateral displacement is strictly below the allowable limit. Therefore, the close-up photographs will maintain proper alignment.

Heating of a Pump Barrel and Bursting of a Volleyball

Why the pump barrel became hot:

- This is due to adiabatic compression and the thermodynamic work done on the system.
- When air is rapidly pumped into the ball, mechanical work is done directly on the air molecules, which significantly increases their internal energy and raises the temperature. The metal barrel conducts this thermal energy, making it hot to the touch.

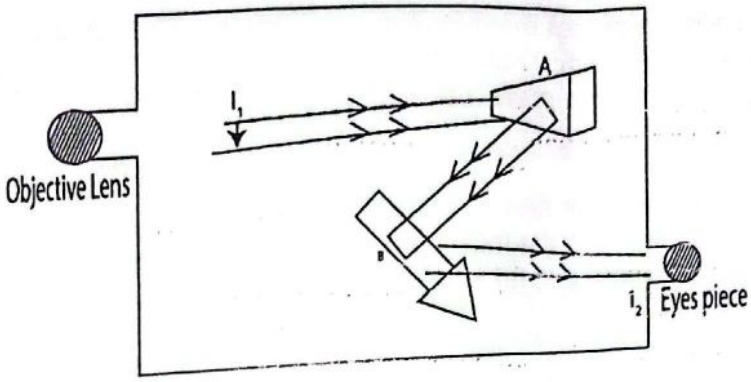
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Why the volleyball burst in hot sunshine:

- In hot sunshine, the temperature of the air trapped inside the ball increases.
- This causes an increase in the average kinetic energy and rate of collisions of the air molecules, leading to a sharp rise in internal pressure.
- Simultaneously, the structural rubber material becomes structurally weaker and more pliable when exposed to intense heat. When the rising internal pressure exceeds the structural strength limit of the rubber, the ball bursts.

b) Optical Instrument Recommendation

Recommendation: Use Binoculars or a Refraction Telescope to effectively magnify distant athletes.



Principles of Operation:

- The objective lens refracts light rays coming from a distant object to form a real, inverted image (I_1) at its principal focus in front of Prism A.
- Rays from I_1 incident on Prism A are totally internally reflected and turned around horizontally by the prism structure.
- On reaching Prism B, the rays undergo total internal reflection again but are flipped vertically, forming a final erect image (I_2) right-side up relative to the distant object, viewed through the eyepiece at infinity.

Item 4

- A concave mirror reflects parallel incoming solar rays precisely onto its focal point.
- The concave mirror tracking alignment is altered to face directly towards the sun.
- A blackened cooking pot is moved incrementally along the principal axis of the mirror layout.
- The positions correspond to an optimal focal setup adjusted until the reflected light forms the smallest and brightest concentrated spot directly on the container.
- The blackened cooking pot absorbs the radiation reflected from the mirror determining If the system is efficient for use.

solar power incident on the mirror:

$$P = IA \ , \ A = \pi r^2_m \ ; \ r_m = 1.25 \text{ m}$$

$$P = 227 \times \pi \times (1.25)^2 = 1359.721 \text{ W}$$

40% reaches the blackened cooking pot:

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$$P_0 = 0.4 \times 1359.721 = 543.89\ W$$

Heat needed to raise the temperature for water to be used in biology practical:

$$543.89 \times 27 \times 60 = 3.5 \times 4200 \times (\theta - 20)$$

$$\theta = 79.74°C\ (= 80°C)$$

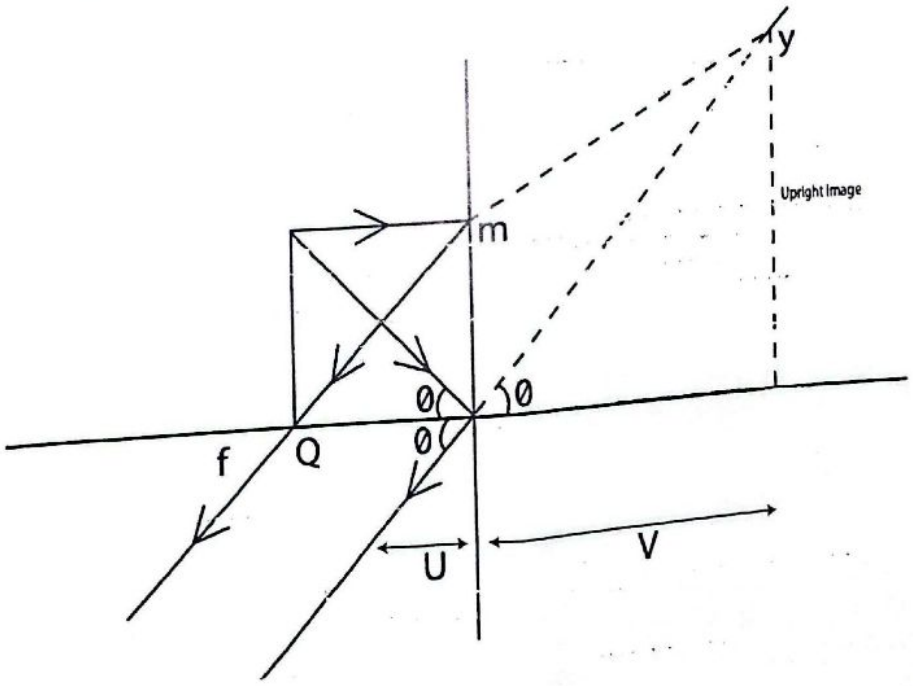
This matches the precise required baseline temperature needed for the biology practice.

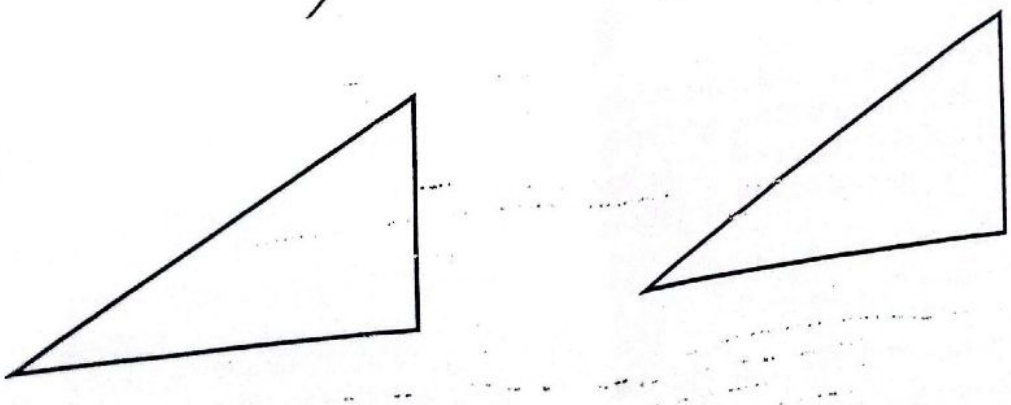
Technical Enhancements to Achieve Boiling Point (100°C)

- Increase effective area captures more solar power.
- Use a transparent cover creates greenhouse effect, reduces convection and radiation loses
- Add reflective side walls: Increases the effective collection aperture area.
- Use an active tracking mechanism: Keeps the reflector pointed at the sun for longer periods, maximizing daily average power output.

b)

- A dentist requires a larger, clearer view of a small, hard-to-access tooth surface.
- When the tooth is placed between the focal point (F) and the pole (P) of a concave mirror, the image becomes enlarged, upright and virtual





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$\frac{yl}{mp} = \frac{u+v}{f}$ since ox = mp

$\frac{yl}{ox} = \frac{u-v}{f}$

$=> \frac{-v}{u} = \frac{-v+u}{F}$, $\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$

Item-5

a) why electricity transmitted at 135KV but suppiied to homes at about 240V

- Electrical power is transmitted at very high V=135KV, so that the current is small
- since P=VI for the same power, increasing voltage reduces current and power lost in transmission lines is Ploss = I^2R, the smaller current greatly reduces heat and energy losses in transmission wires

Why are bluish glows and faint sparks

$P=VI$, $I = \frac{5X10^6}{135000} = 37.04A$

Since 37.04A < 40A bluish faint light is caused by a iarge current flowing in wires

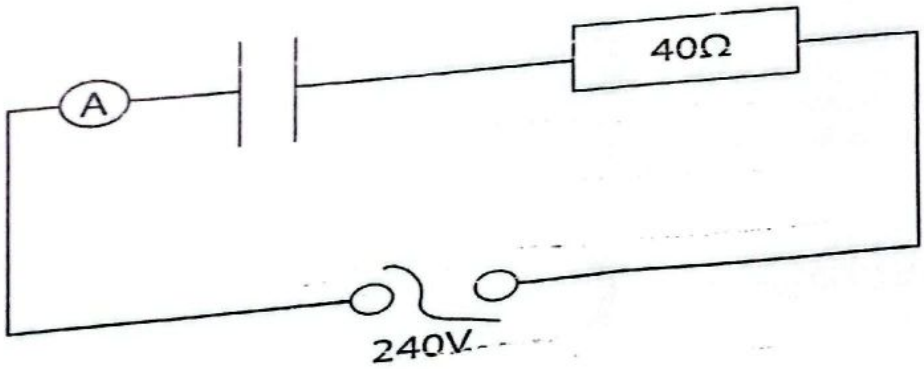
the glow is due to Corona discharge the high voltage produces a very high electric field intensity around the conductor

Air molecules near the wire become ionized, the ionized air emits a faint bluish light and sometimes produces a hissing sound not caused by a large current flowing in wires since transmission current is limited < 40A, power loss is kept very low.

Concern of health center technician

$\frac{V_S}{V_P} = \frac{N_S}{N_P}$, $V_S = \frac{240 \times 25}{500} = 12V$

Transformer appropriate because it steps down the main supply from 240V to 12V medical monitoring circuit is to be connected to a fuse device, 40A or 6.0A



$V = ZI$

$Z = \sqrt{X\frac{2}{C} + R^2)}$, $Xc = \frac{1}{2\pi fc}$

$Xc = \frac{1}{2\pi \times 106 \times 10^{-6} \times 50} = 30.029\Omega$

$Z = \sqrt{(30.029)^2 + 40^2} = 50.018\Omega$

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$240 = 50.018\,I,\ I = 4.80A$

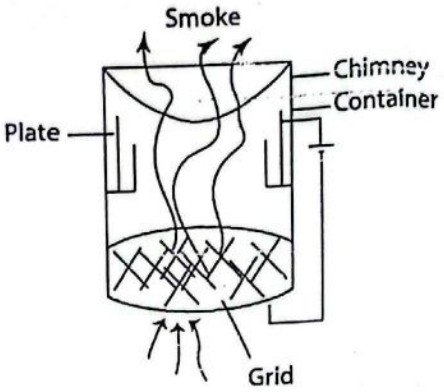
40A fuse; Too small it will blow immediately because 4.80A > 4.0A

Page 12 (First Sheet)

6.0A fuse; can carry 4.80A safely and will not blow if there is a fault / overheating
∴ The 6.0A fuse is most appropriate

b) Recommend an electrostatic

- based solution; electrostatic air precipitation.



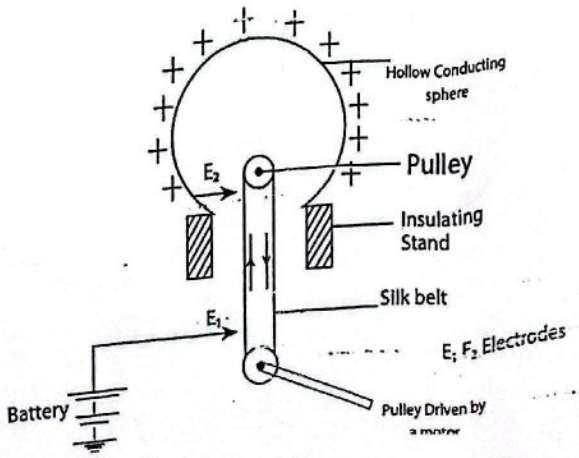
- In a chimney there is a grid which is highly negatively charged with respect to the plates
- As smoke passes through the grid its particles are negatively charged
- when the smoke reaches the regions of plates, the particles are attracted to the plates
- Hence smoke gets out of the chimney free of particles
- The particles accumulating on the plates falls off in the container when heavy enough.

Advantages

- Reduces respiratory health risks to patients and staff
- Improves air quality in the intensive care unit.
- Has high efficiency
- Removes very fine dust particles

item 6

Work of a van de Graaff generator



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- Electrode E_1 is made positive relative to the Earth, at E_1 there is high charge density and high electric field intensity. Air around E_1 gets ionized
- Positive ions are repelled to the belt which drives them up to the sphere
- At the sphere negative charge is induced at E_2 and positive charge induced at the surface of the sphere.
- Point action takes place at E_2, negative ions are repelled to the belt and discharge it before it moves over the pulley.
- The process continuous until the charge on the sphere can ionize the air around it.

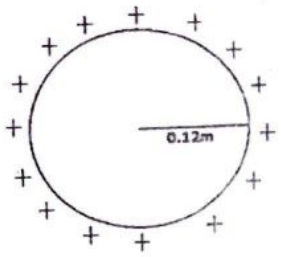
Decision to accept the generator or not,

Energy supplied by the motor, E = V I t

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$E = 12 \times 0.5 \times 60 = 360J$

Energy stored by the sphere.



$Q = 2.4 \times 10^{-6}\ C$

$C = 4\pi\ \varepsilon_0\ r$

$C = 4\pi\ (8.85 \times 10^{-12}) \times 0.12$

$C = 1.33 \times 10^{-11}\ F$

Electrostatic energy

$$E = \frac{Q^2}{2C}$$

$$E = \frac{(2.4 \times 10^{-6})^2}{2 \times 1.33 \times 10^{-11}} = 0.216J$$

Efficiency of conversion of energy

$$\eta = \frac{0.216}{360} \times 100\% = 0.06\%$$

Required efficiency of energy conversion is 0.06% > 0.05%

For the van de Graaff generator to be accepted, the sphere must be smooth, spherical and clean, thus charge must be uniformly distributed over the surface to check for this

- A proof plane is placed on the surface of the dome that is charged, and then transferred into a metal can placed on the cap of a neutral gold leaf electroscope
- Divergence on the gold leaf electroscope is noted.

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- The experiment is repeated with the proof plane placed at different parts of the dome and corresponding divergencies noted
- If the divergencies are equal, the charge is then uniformly distributed since efficiency of energy conversion is greater than 0.05%, and charge distributed uniformly over the dome, the generator can be accepted.

b) - As the van de Graaf is kept running charge on the dome kept on increasing until a value when the insulation of air breaks down.

since $Q = E_0 \times 4\pi \varepsilon_0 r^2$

$Q = 3.0 \times 10^6 \times 4\pi (8.85 \times 10^{-12}) (0.12)^2$

$Q = 4.8 \times 10^{-6} C$

When the charge increases to 4.8×10^{-6} the electric field intensity is high enough to break down insulation of the air around the sphere thus causing sparks around the dome.

END

ex — Explanation (Knowledge, description of expt, effects)
AN — Analysis (All numerical computation) + a conclusion
AP — Application (Reallife applications e.g devices) — graphs

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